

LEVEL 1 · PARTICIPANT REFERENCE

Glossary for Modules 1 & 2

GenAI & LLM Fundamentals for Testers

How AI Applications Are Built

A post-session reference for key terms, concepts and testing language

How to use this glossary

You do not need to memorise every term. Use this as a reference after the session. When analysing an AI defect, try to describe what actually happened - for example, “the wrong document was retrieved” rather than simply saying “the AI hallucinated”.

Module 1 - GenAI & LLM Fundamentals for Testers

Use these terms when describing model behaviour, variation, settings and test evidence.

Term	What it means and why it matters
Artificial Intelligence (AI)	A broad term for systems that perform tasks associated with human intelligence, such as making decisions, recognising patterns, understanding language or giving recommendations. AI includes machine learning, deep learning and generative AI.
Machine Learning (ML)	A type of AI in which the system learns patterns from data instead of relying only on rules explicitly written by developers. Examples include fraud prediction, recommendation systems and predictive-maintenance models.
Deep Learning	A type of machine learning that uses large neural networks and is commonly used for images, speech and language. At Level 1, the important point is not the mathematics, but that behaviour is learned rather than expressed as simple business rules.
Generative AI (GenAI)	AI that creates new content such as text, images, code, summaries or test data. Because the content is generated, more than one response may be acceptable for the same input.
Large Language Model (LLM)	A generative AI model designed to work with language. An LLM produces text by repeatedly predicting a likely next piece of text based on the prompt and context it receives.
Prompt	The instructions or input given to an LLM. A prompt influences what the model is expected to do, but it is not a hard guarantee that every instruction will always be followed perfectly.
Token	A piece of text processed by the model. A token may be a whole word, part of a word or punctuation. Both input and output are measured in tokens, so tokens influence cost, speed and model limits.
Context	Everything available to the model for the current response - system instructions, user question, conversation history and any supplied evidence. If the required information is not in the context, the model cannot reliably use it.
Context Window	The maximum amount of information a model can consider in one request. Very large or noisy contexts can also make important details easier for the model to overlook, even when they technically fit.
Next-token prediction	The basic behaviour behind LLM generation: the model estimates likely next tokens, chooses one, adds it to the text and repeats. This helps explain why fluent output can still be wrong and why repeated runs can differ.
Probabilistic Output	LLMs do not always make exactly the same token choice. Therefore, the same prompt can produce somewhat different answers across runs. Variation itself is not automatically a defect.
Non-determinism / Variance	The variation seen when the same or equivalent input produces different outputs. Testers respond by repeating important tests and checking whether required behaviour remains stable across runs.
Test Oracle	The basis used to decide whether a test passes or fails. Traditional systems often have one exact expected value; AI-generated responses usually need a set of acceptance criteria rather than one fixed sentence.

Term	What it means and why it matters
Acceptance Criteria	Observable conditions an AI response must satisfy. Examples: must contain the correct value, must include a safety warning, must not invent information, and must stay within a requested format.
Temperature	A generation setting that affects how adventurous token selection is. Lower temperature generally gives more similar responses; higher temperature generally allows more variety. Lower temperature does not make an answer automatically correct.
Top P	A generation setting that restricts the pool of candidate tokens from which the model can choose. When studying temperature, testers normally keep Top P unchanged so the effect of temperature can be isolated.
Controlled Experiment	A test in which only one variable is deliberately changed at a time. For example, when studying temperature, keep the model, prompt, Top P and output limit fixed.
Model	The underlying AI system that performs the generation. Different models can differ in instruction following, speed, cost, reasoning ability, safety behaviour and consistency.
Model Version	A particular release of a model. A provider update can change behaviour even when the application code has not changed, so a model-version change is a regression trigger.
Model Comparison	Testing two models on the same task with the same prompt, settings and agreed criteria. The objective is to determine task fit, not to declare one model universally better.
Regression Testing for AI	Re-running important checks after something changes, such as a model version, prompt, generation setting or later a knowledge document. AI behaviour can change without a normal code change.
Repeated-run Testing	Running the same test several times under the same conditions. This helps expose intermittent failures, such as a required warning being omitted in one out of five runs.
Unsupported Claim	Information stated by the model without evidence available in the prompt or approved context. A claim can sound sensible and confident while still being unsupported.
Hallucination	A commonly used term for fabricated, incorrect or unsupported generated information. Testers should prefer a more precise failure description when possible, such as unsupported claim or groundedness failure.
AI Defect Evidence	The details needed to investigate an AI failure: exact prompt, model/version, settings, full response, number of runs and failure frequency. These details make intermittent AI behaviour easier to reproduce and diagnose.

Module 2 - How AI Applications Are Built

Use these terms when tracing a response through documents, retrieval, generation, tools and output.

Term	What it means and why it matters
AI Application / AI Feature	The complete system around the model. It can include the user request, system instructions, context, document retrieval, the LLM, tools, output handling and application logic. A wrong answer does not automatically mean the model itself failed.
System Instruction / System Prompt	Instructions defined by the application team that tell the model its role, boundaries and expected behaviour. Example: "Answer only from the uploaded manuals and do not guess."
Context in a RAG Application	The evidence and instructions supplied to the LLM for the current request. Retrieved document extracts become part of this context before the model generates the answer.
RAG (Retrieval-Augmented Generation)	A pattern where the application searches for relevant information first and then gives that evidence to the LLM before it answers. In simple terms: search first, answer second.
Grounding	Connecting an AI response to approved evidence such as a manual, policy or database record. A grounded answer should be supported by the evidence supplied to the model rather than unsupported general knowledge.
Fine-tuning / Retraining	A process that changes the model itself using additional training. It is different from RAG: adding or updating RAG documents changes what the model is shown, not the base model.
Document Ingestion / Preparation	The processing that happens when a document is added to a RAG system. Text may be extracted, cleaned, split into chunks, converted into embeddings and stored in a searchable index.
Parsing / Text Extraction	Reading usable text from a PDF, Word file or another source. If a table, image or scanned page is not extracted correctly, the information may never become searchable.
Chunk	A smaller piece of a larger document. RAG systems retrieve relevant chunks instead of sending an entire large document to the model for every question.
Chunking	The process of dividing documents into chunks. Poor chunking can split related facts or remove necessary surrounding context, which can cause retrieval failures later.
Embedding	A numerical representation of the meaning of text. Embeddings allow the system to compare the meaning of a user question with the meaning of document chunks.
Embedding Model	The model used to create embeddings. Its job is to support semantic retrieval; it normally does not generate the final natural-language answer.
Vector Search	Search based on semantic similarity between embeddings rather than only exact keyword matches. This allows differently worded questions with the same meaning to retrieve relevant content.
Index	The searchable store containing prepared document information and embeddings. The retrieval component searches this index when a user asks a question.

Term	What it means and why it matters
Retrieval	The process of finding document chunks that appear most relevant to the user's question. Those chunks are supplied to the LLM as evidence for generation.
Retriever	The component that performs the search and selects the best-matching chunks. If it selects the wrong product manual, the LLM may receive the wrong evidence.
Top Retrieved Chunks / Top Results	The document pieces ranked as the best matches for a question. Testers should inspect whether the correct evidence actually appears among these results.
Generation	The stage where the LLM uses the prompt and retrieved evidence to produce the final answer. Retrieval and generation are separate stages and can fail independently.
Retrieval Failure	The correct evidence is missing from what reaches the LLM. Examples include retrieving the wrong product manual or failing to retrieve the section that contains the required answer.
Generation / Groundedness Failure	The correct evidence was retrieved, but the final answer ignores, misreads or contradicts it. Example: the retrieved text says 10 bar, but the answer says 12 bar.
Compound Failure	A defect where more than one component contributes. For example, the wrong manual ranks first and the model also fails to notice that the evidence belongs to a different product.
Missing-information Test	A test where the knowledge source intentionally does not contain the requested answer. A well-grounded assistant should state that the information is unavailable rather than inventing a plausible answer.
Paraphrase Test	Asking the same question in different words. It checks whether semantic retrieval reaches the same evidence and whether the substantive answer remains consistent.
Leading-question Test	A test where the user suggests an incorrect answer or assumption. A robust assistant should stay with the evidence instead of agreeing merely because the user sounds confident.
Citation	A reference to a source used for an answer. A citation is useful for traceability, but its presence alone is not proof - the cited passage must actually support the claim.
Trace	A record showing what happened inside the AI application for a request, such as tool calls, retrieval steps, returned evidence and token usage. It plays a similar role to logs in traditional debugging.
Tool	An external capability the AI system can use, such as document search, a calculator, database access or creating a service request. Tools extend the model beyond simply generating text.
Tool Call	A request from the AI system to execute a tool, usually with parameters. Testers should verify the right tool was selected and that its parameters and resulting action are correct.
Agent	An AI system that can decide which steps to take and which tools to call to achieve a goal. An agent can therefore do more than simply retrieve information and generate an answer.
Agentic AI	AI behaviour where the system can plan or choose actions and interact with external tools or systems. Risk increases because an incorrect decision may create a real-world action rather than only a wrong sentence.

Term	What it means and why it matters
Suggesting vs Acting	A RAG assistant might say, “You should create a service request.” An agent may actually call a tool and create one. Acting requires extra testing of authorization, parameters, confirmations and audit evidence.
Output Handling	What the application does with the model response before showing it to the user - formatting, filtering, citations or privacy controls. Retrieval and generation can both be correct while output handling still causes a defect.
Failure-point Register	A testing artefact that lists each component, what could fail, what evidence would expose the failure, its severity and possibly the likely fix area. It helps teams avoid treating the whole solution as one “AI box”.

One habit to take away

When an AI response is wrong, avoid stopping at “the AI gave a wrong answer”. Ask what evidence you have: Was the prompt clear? Was the right context present? Did retrieval find the right source? Did generation use it correctly? Did a tool take the right action? This language makes defects easier to diagnose and fix.